

Shantanu Shastri

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EDUCATION

Carnegie Mellon University	December 2026
Master of Information Systems Management	GPA: 3.85
Manipal Institute of Technology	August 2022
Bachelor of Electronics and Communications Engineering	GPA: 3.99
Relevant Coursework: Deep Learning, Generative AI Lab, LLM Applications, MLOps, System Design	

SKILLS

Languages: C++, Python, JavaScript, Golang, TypeScript, Java, SQL, Bash/Shell
Frontend & APIs: React, GraphQL, API Layers, TypeScript Micro-frontends, UI Design, Node.js
Infrastructure: Docker, Kubernetes, Containerization, Cluster Orchestration, Apache Kafka, Zookeeper, CI/CD
Full-Stack Engineering: System Architecture, Database Optimization, Microservices, Spring Boot, PostgreSQL

EXPERIENCE

BlackRock	January 2022 – August 2026
AI Engineer Internship	June 2026 – August 2026

- Complex Software & Tools:** Engineered complex software across the entire stack to develop new features and tools, building a multi-agent platform (RockAI ADK) that automated sophisticated workflows.
- Infrastructure & Data:** Designed efficient and effective solutions for infrastructure challenges by linking Milvus embeddings to Neo4j, enabling secure, structured search with sub-100ms latency.
- Ownership & Feedback:** Took ownership over technical decisions and closely interacted with users to collect feedback, implementing offline-online evaluation frameworks to process 3.5M records reliably.

Software Developer 2 - Associate	January 2025 – July 2025
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- Frontend & API Layers:** Architected and shipped TypeScript/React micro-frontends and robust API layers to translate natural language into JQL, acting as a self-starter to optimize and streamline user workflows.
- Full-Stack Microservices:** Engineered scalable Spring Boot data frameworks and TTL Azure microservices from scratch, managing complex dependencies and reducing infrastructure storage costs by 40%.

Software Developer 1 - Analyst	July 2022 – December 2024
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- Cluster Orchestration:** Built comprehensive infrastructure by engineering a sharded server environment with Zookeeper cluster orchestration, boosting system throughput to 10K+ RPS for mission-critical services.
- Data Modeling & Optimization:** Took end-to-end ownership of database optimization, modeling complex data challenges on PostgreSQL tables exceeding 50M+ rows to achieve a 37% reduction in query latency.
- Self-Starter Engineering:** Demonstrated the ability to quickly master new technical tools by building Azure DevOps CI/CD pipelines with integrated JUnit testing suites for robust, full-stack deployments.

Shunya	June 2020 – September 2020
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- Data Science Intern
- Physical Models & Workflows:** Engineered a GPU-accelerated PyTorch CV workflow on a Linux environment, successfully automating complex evaluations and reducing manual grading time by 90%.
 - Data-Driven Solutions:** Addressed learning gaps by training K-means clustering models on multidimensional assessment data, designing efficient solutions that dynamically recommended targeted practice.

PROJECTS

- High-Performance Modular Booking Architecture
- Containerization & Scale:** Engineered a decoupled microservices architecture using Apache Kafka and REST APIs, seamlessly handling bursts of 100,000+ real-time requests with Docker containerization.
- Zulip Open-Source Engine Extensions
- High-Throughput Engines:** Built a robust message engine using Node.js and Redis to process live data streams into concise summaries, prioritizing clean API contracts and zero message loss.